Do Students Agree with LLM Feedback on Their Own Code? A Within-Subject Study of Four Feedback Configurations in Software Project Courses

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Abstract. LLM feedback on explanations of students' own software projects raises distinct questions about agreement, actionability and willingness to use the feedback. We evaluated PACE, a repository-grounded question-and-feedback system, with 86 students from 28 team repositories. Each student rated one instance of each of four assigned configurations: Sandwich, Rubric, Explainable and Researching. These configurations bundle different content requirements and, for Researching, an intended two-stage model pipeline. Across 344 rated instances, 79.9% of ratings endorsed the AI's judgement, 68.0% endorsed future use and 41.9% endorsed motivation. All six 95% repository-bootstrap contrasts on the direct Acceptance item included zero, without establishing equivalence. Sandwich–Explainable actionability differed by 0.37 points (95% CI [0.16, 0.57]); this bundle comparison does not isolate next steps. A secondary multidomain composite averaged 4.07/5 (95% CI [3.90, 4.23]); its small mean differences coexisted with distinct item profiles. An observational raw-code versus graph comparison was inconclusive ($b = 0.07$, 95% CI $[-0.02, 0.20]$). These findings concern reported perceptions, not expert-verified correctness, observed revision or learning gains.

Keywords: LLM-generated feedback · Feedback acceptance · Programming education · Software engineering education · Human–AI trust

1 Introduction

Feedback is one of the strongest levers on learning, but its effect depends on what the learner does with it [12,32]. Technically correct feedback does not by itself demonstrate that students will understand or use it. Research on feedback literacy therefore treats the student's uptake—making sense of a comment, judging it, and deciding to act—as part of the feedback process rather than its aftermath [2,35].

LLMs can identify issues in student code and explain them in readable language, but their comments may also be misleading [14,20,4]. Guardrails can support guidance without supplying complete solutions [26]. Their growing role in computing education [6,17] makes student acceptance worth examining separately from technical quality. Evidence varies by setting: an ENL writing study found split preferences between tutor and AI

feedback [10], while programming studies motivate examining specificity, provenance and trust separately.

Two gaps motivate this study. First, project courses assess evolving team repositories that no student wrote alone. Questions about learners' own programs can probe comprehension [25], but perceptions of LLM feedback on those answers remain underexplored in the literature discussed here. Second, deployed systems bundle praise, criteria, evidence and model orchestration into different configurations, each with a pedagogical rationale [29,16,19,8]. We compare students' ratings of these bundles without assuming identical underlying judgements.

We address these gaps with PACE (Project Assistance and Course Evaluation), a platform deployed in software project courses at FPT University. PACE analyses each team's repository, generates scenario questions about business rules implemented in it, and returns LLM feedback on every answer. Each student answered four questions and received feedback in four different configurations, assigned by the described permutation procedure, then rated every feedback instance on fifteen items covering perceived quality, utility, trust, affect, and acceptance. We ask:

RQ1 How do students judge the quality of LLM feedback on their answers about their own code, and how far do they accept it?
RQ2 Does student agreement differ across the four deployed feedback configurations?
RQ3 How is student agreement associated with raw-code versus graph-based question contexts in the deployed system?

Our contribution is a field evaluation of feedback on explanations of business rules in shared team repositories. It extends questions about learners' code to a project-defence preparation workflow and separates agreement with a judgement from its perceived usefulness, motivation and intended use. This is an integration and empirical-evaluation contribution, not a new LLM algorithm. The evaluation concerns reported perceptions of complete deployed configurations; it does not establish improved learning or superiority to instructor feedback.

2 Related Work

2.1 LLM-Generated Feedback in Programming Education

Automated feedback predates LLMs: a review of 101 programming tools found more emphasis on identifying mistakes than on supporting next steps [18]. LLM evaluations report useful issue identification alongside missed or invented issues [14,20,4]; CodeHelp demonstrates guarded course-scale assistance [26]. Our task concerns feedback on explanations of team-repository behaviour, rather than feedback on an isolated programming solution.

2.2 Student Perceptions and Acceptance of AI Feedback

Usefulness and trust are related to intentions to continue using a system [5,34], while appropriate reliance requires calibration to actual reliability [23]. Algorithm aversion [7]

and appreciation [27] both occur. In programming education, Zhang et al. reported preferences for code-referencing feedback [36]; Er et al. found advantages for instructor feedback [9]; and Nazaretsky et al. found perceptions sensitive to AI provenance [28]. These findings motivate measuring trust and intention alongside perceived quality.

2.3 Feedback Configurations

Sandwich feedback improved perceptions without improving performance in a controlled study [29]. Rubrics make criteria explicit [16]. Evidence-based explanations may support appropriate trust [19]. Multiagent debate has improved factuality on benchmark tasks [8], but our sequential inspection–synthesis pipeline does not implement debate. Model-as-judge benchmarks also document position and verbosity biases alongside agreement with human preferences [37]. Neither result validates PACE feedback against expert judgements. Our within-student comparison evaluates complete bundles.

Position relative to the closest studies. Questions about learners' code already motivate automatic comprehension checks [25]; our distinction is feedback on explanations of business rules across a shared team repository. Zhang et al. studied perceived formative feedback for CS1 Java assignments and compared prompts with and without code [36]; PACE instead compares four complete feedback configurations on comprehension answers. Lehtinen et al. studied LLM answers to comprehension questions about model-generated programs [24]; that work motivates checking reference-answer fallibility, whereas our outcomes are student-reported perceptions. These differences locate the contribution in task and evaluation context, rather than a new question-generation or feedback algorithm.

3 The PACE System

Figure 1 summarises repository analysis, question generation and feedback generation in PACE.

3.1 Repository Analysis and Question Generation

For each team repository, PACE clones the submitted branch and parses it with Roslyn (C#) or tree-sitter (Java, JavaScript and TypeScript). Endpoints are grouped into *business functions* (e.g., "contract management" or "payment confirmation"), either from a template stored in the repository or from a draft that an instructor reviews and confirms. For every function, PACE resolves calls from each entry point and stores both the source code of the involved methods and a data-flow graph with ordered call edges, branch conditions and a Mermaid rendering.

Questions are generated per business function by the configured question-generation LLM (temperature 0.3, structured JSON output), which receives either the raw source code or the Mermaid graph; a router selects the graph when the flow involves more than ten distinct methods and the raw code otherwise. The prompt targets scenario questions about guard conditions and failure branches, for example: *"When the facility manager*

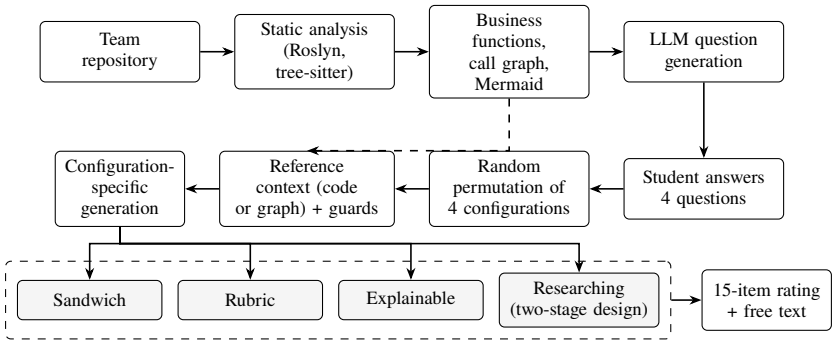


Fig. 1. Overview of PACE. Questions and the evaluator's reference context are both derived from the student's own repository. Each student's four answers receive the four configurations in a random order.

tries to swap the storage unit of a contract that still has an outstanding balance, how does the system respond and why is the action blocked?" Each question is stored with a suggested answer and with the context type used to produce it.

3.2 Feedback Generation

When a student submits answers, each answer is evaluated against a *reference context* built from the same business function: the raw source of its methods, or its flow graph with numbered steps and conditions, following the context type of the question. The system prompt fixes an instruction hierarchy: content inside the `<student_answer>` tag is data, never instructions; the reference context and the suggested answer supply the evaluation reference, whose correctness is not independently established here; the model comments and explains but does not assign an overall grade; and the output must be a single JSON object. An input guardrail rejects answers that contain prompt-injection phrases or executable payloads, and an output guardrail validates the JSON before it is stored.

The four configurations bundle different content requirements and generation procedures:

- **Sandwich**: positives, constructive criticism, and next steps.
- **Rubric**: a list of criteria that the model names for the question (e.g., understanding of the business flow, identification of the guard condition), each with a 1–5 score and a comment.
- **Explainable**: a *code evidence* field that cites methods, steps or conditions from the context, and a *logic explanation* that relates them to the answer.
- **Researching**: a two-stage design: a DeepSeek model checks the answer against the code, then GPT-OSS-120B (served by Groq) synthesises a report: an overall comment, positives, criticism, next steps, code evidence, two scored criteria, and a consensus summary with pedagogical guidance.

For the contract-balance question in the preceding subsection, the common evaluation task is whether the answer correctly explains the blocking condition and its consequence. Sandwich requests positives, criticism and a revision step; Rubric requests criterion scores and comments; Explainable requests contextual evidence and reasoning; Researching requests an inspector result followed by synthesis. This is a schematic explanation of the content requirements, not four observed outputs from the same input. The study used different answers for each configuration, so identical judgements across configurations are not assumed.

Only Sandwich and Researching are designed to require an explicit list of next steps. The three single-model configurations specify `deepseek-chat` (temperature 0.2, JSON mode). This API alias does not identify an immutable model version; reproducibility requires provider IDs, deployment revision and execution dates. If the second model in the Researching pipeline fails, the system falls back to the single-model path; all 86 Researching records list both roles in nested metadata (DeepSeek-V3 and Groq-GPT-OSS-120B). This supports the recorded two-model configuration; independent provider execution logs are not part of the export.

Researching is the configuration name; its consensus field is a generated summary, not independent expert agreement. The assigned configuration is the intended experimental manipulation; exported labels establish complete blocks but do not independently authenticate the randomisation implementation. Comparisons use assigned labels, including any system guardrail responses. When a student submits $n = 4$ answers, PACE shuffles the four configurations and assigns one to each answer, so every student sees every configuration exactly once, in random order.

4 Study Design

4.1 Context and Participants

The study took place in October 2026 in undergraduate software engineering project courses at FPT University, Vietnam. The research team visited classes and invited students to use PACE and provide candid feedback as preparation for questions they might encounter before a project defence panel. Students used the system to rehearse explaining their team's repository. Classes were selected at random, and participation within each class was voluntary: some students participated and others declined. The class sampling frame, procedure, invitation count and response rate remain undocumented; respondents are self-selected within classes. The sample comprises 86 students from 28 team repositories, answering 190 distinct questions on 152 business functions. Answers were written in Vietnamese and were short, about 51 whitespace-separated tokens on average. Submissions were made between 20 and 26 October 2026.

4.2 Instrument

After reading each feedback instance, students rated it on fifteen statements on a five-point scale (1 = strongly disagree, 5 = strongly agree) and could add a free-text reply (160 non-empty fields). No independently coded qualitative prevalence or observed uptake is inferred from these fields. The items were written in Vietnamese by the research

Table 1. Survey instrument (translated from Vietnamese). †Reverse-worded.

Construct	Item	Statement
Perceived quality	Accuracy	The AI's comment correctly identifies the problems in my answer.
	Specificity	The AI points to exactly which part of my answer needs revisiting, rather than commenting in general.
	Clarity	The wording and presentation of the comment are clear and unambiguous.
	Transparency	I understand which academic criteria the AI used to assess my answer.
Perceived utility	Usefulness	This AI tool is an effective learning assistant for my project work.
	Actionability	I know exactly which technical steps to take to improve my answer.
Trust	Competence	I trust the technical expertise and analytical capability of this AI.
	Reliability	The AI assesses consistently and does not change its standards at random.
	Fairness	The AI's assessment fairly reflects the effort and actual quality of my answer.
Affect	Motivation	I feel motivated to keep learning after reading the AI's comments.
	Psych. safety	AI feedback causes less pressure than being criticised by a person.
Acceptance & intention	Acceptance	I agree with the AI's judgement of my answer's strengths and weaknesses.
	Willingness	Based on this comment, I am ready to revise my answer right away.
	Behav. intention	I intend to keep using this LLM system for feedback in future projects.
Separate item	Rejection†	I tend to want to refute or argue against the AI's assessment.

team, drawing on formative-feedback principles [32] and technology-acceptance constructs [5,34,23]. Table 1 lists them with English translations, grouped into five a priori constructs. REJECTION is reverse-worded.

4.3 Data Set

The survey export contained 344 answer records from 86 submissions, one submission per student. The research team confirms voluntary student completion of all 344 records; analysis checks the export, not database history. Screening checked stored feedback, complete ratings and explicit development-fixture markers in feedback metadata; all 344 records passed. Feedback-origin metadata concerns the AI output, and does not independently authenticate the origin of survey ratings. Every student rated all four configurations, which gives 86 complete blocks and 86 instances per configuration. One assigned Rubric instance contains a guardrail refusal with next steps but no criteria; the other 85 contain criteria. We retain assigned labels rather than reclassifying outputs. The export also stores the context type of each question: 280 instances (157 questions) came from graph-based questions and 64 instances (33 questions, 13 repositories) from raw-code questions. Thirty-two students received at least one raw-code question, and 26 of them answered questions of both types.

4.4 Analysis

Scores. The *AI agreement index* (AAI) is an exploratory composite: the mean of fourteen non-reversed items across several domains. It is not a validated unidimensional measure of agreement. The Acceptance item is the direct outcome for agreement with the AI judgement; motivation, willingness to revise and future-use intention are distinct outcomes. Construct and item profiles are the main descriptive evidence. AAI is

a secondary multidomain summary with equal weight per item, so domains with more items receive greater total weight. This hierarchy is a revision choice, not a preregistered endpoint specification. Excluding Rejection was data-informed (Section 5.1) [33].

Reliability and data quality. We report Cronbach's α [3] and McDonald's ω_t from a one-factor solution [13], from the pooled repeated-instance covariance, without interpreting either as within-student reliability or factorial validation, and flag *straight-lined* responses—identical ratings on all fifteen items—following Kim et al. [21].

RQ1. We report item means, distributions and 95% percentile confidence intervals from resampling all 28 repositories with replacement (5,000 replicates; seed 20261029). Each sampled repository contributes all its students and questions; differences are recomputed within the same resampled data, rather than inferred from overlap of marginal intervals; means weight feedback instances equally. These intervals allow dependence within repositories, assuming independence across repositories [1]. To explore conditional associations with acceptance, we fit student fixed-effects regressions after demeaning predictors and outcomes within each student:

$$y_{ij} - \bar{y}_j = \sum_{k=1}^{5} \beta_k (z_{kij} - \bar{z}_{kj}) + \sum_{c=2}^{4} \gamma_c (D_{cij} - \bar{D}_{cj}) + e_{ij}, \tag{1}$$

Here i indexes feedback instances, j students, y is Acceptance, Willingness or Behavioural intention, and D_c are configuration indicators (Sandwich reference). Predictors are perceived quality, perceived utility, trust, motivation and psychological safety, standardised using their full-sample SDs before demeaning. Motivation and psychological safety remain separate because their statements concern different experiences; pooled reliability in this export is also low. Regression intervals use the same repository bootstrap. Resampling retains each complete student block; repeated draws of a repository repeat its contribution without splitting the block. All 5,000 archived fixed-effects bootstrap fits were full rank. Predictor standardisation is fixed at the original full-sample SDs across resamples. These models describe contemporaneous within-student associations, not causal drivers or differences between students.

RQ2. The six paired Acceptance contrasts are the principal configuration comparisons; AAI contrasts are secondary. We report first-named minus second-named paired means and repository-bootstrap intervals. Exploratory Friedman tests [11], Kendall's W, Holm-adjusted item/construct and Wilcoxon tests [15], and student-intercept mixed models are retained as secondary analyses because they do not fully model shared-repository dependence. A length-adjusted mixed model additionally includes feedback and answer length. For AAI only, we assess whether paired 90% intervals fall inside ± 0.25, with ± 0.20 and ± 0.10 sensitivity checks [22]. These exploratory margins lack stakeholder or learning-outcome calibration. Scoring sensitivity reverse-scores Rejection and retains all fifteen items; response-pattern sensitivity excludes complete blocks affected by straight-lining or the guardrail refusal.

RQ3. Context type was not randomised: the router uses graph context for functions involving more than ten methods and raw code otherwise. Complexity, question context and evaluator context are therefore coupled. We report descriptive means and an exploratory student fixed-effects model with context and configuration indicators, using repository-bootstrap intervals. The context coefficient uses within-student variation, supplied by 26 students exposed to both types. It remains an association confounded by complexity and is not an isolated context effect.

Dependence and scope. Answers are assigned within students, who share repositories and questions. Repository resampling preserves students and shared questions: each belongs to one repository. Rank tests and student-intercept mixed models do not fully capture repository dependence; substantive comparisons therefore emphasise repository-bootstrap intervals. We also delete each entire repository in turn and recompute paired means and the context coefficient. These point-estimate ranges diagnose single-cluster sensitivity, not sampling uncertainty. Straight-lining is a response-pattern flag, not proof of inattention.

Analyses use Python and statsmodels [31]. Analysis scripts and aggregate artifacts are retained in the local research workspace; a reviewer-accessible supplement has not yet been released. Identifier-bearing raw exports are withheld. Release of an anonymised dataset requires an appropriate sharing basis.

5 Results

5.1 Reliability and Data Quality

The fourteen non-reversed items had high internal consistency ($\alpha = .936$, $\omega_t = .939$). Item-group estimates were $\alpha = .91$ for perceived quality, .78 for utility, .80 for trust and .78 for acceptance & intention. Affect had lower consistency (.56): its two items, motivation and psychological safety, should not be assumed to form a single reliable construct. Internal consistency alone does not establish dimensionality or construct validity. Two observations qualify this.

There are 64 straight-lined instances (18.6%) and 11 students who straight-lined all four ratings (12.8%). Excluding flagged rows gives $\alpha = .90$, but does not identify attentive respondents. Rejection's reverse-scored item–rest correlation is $-.03$; its descriptive correlation with Acceptance is $-.14$, versus $-.40$ after excluding straight-lined rows. Reverse wording may produce inconsistent responses [33]; excluding this item remains a data-informed choice. These pooled response-pattern statistics do not establish psychometric validity or justify independent-row inference.

5.2 RQ1: Item Profile in the Export

Table 2 describes ratings across the 344 feedback instances. Ratings exceed the neutral midpoint on every non-reversed item. The agreement index averaged 4.07 (SD 0.64, 95% repository-bootstrap CI [3.90, 4.23]). Students most often endorsed usefulness (84.0% agree), accuracy and specificity (83.4% each), and explicit disagreement stayed at or

Table 2. Item ratings across 344 feedback instances. CI: 95% repository-bootstrap interval of the mean (28 repositories). Agree/Disagree: % of ratings ≥ 4 / ≤ 2. †Reverse-worded, not reverse-scored.

Item	Mean (SD)	95% CI	Agree	Disagree
Accuracy	4.38 (0.96)	[4.14, 4.61]	83.4	5.5
Specificity	4.36 (0.94)	[4.10, 4.60]	83.4	5.2
Clarity	4.33 (0.88)	[4.09, 4.56]	81.7	3.5
Transparency	4.29 (0.95)	[4.03, 4.54]	79.9	4.9
Usefulness	4.09 (0.77)	[3.93, 4.24]	84.0	2.9
Actionability	4.13 (0.88)	[3.90, 4.34]	80.8	4.9
Competence trust	4.17 (0.98)	[3.94, 4.41]	76.5	4.9
Reliability	3.88 (0.80)	[3.73, 4.01]	78.5	4.9
Fairness	4.15 (0.79)	[3.98, 4.31]	81.4	2.3
Motivation	3.53 (0.87)	[3.38, 3.71]	41.9	4.9
Psych. safety	3.83 (0.85)	[3.65, 3.99]	76.2	7.3
Acceptance	4.09 (0.82)	[3.90, 4.27]	79.9	3.5
Rejection†	2.87 (1.22)	[2.56, 3.20]	30.8	48.8
Willingness	3.97 (0.69)	[3.86, 4.08]	81.4	2.3
Behav. intention	3.80 (0.85)	[3.65, 3.94]	68.0	5.2

Table 3. Student fixed-effects regressions (Eq. 1). Cells show rating points per full-sample predictor SD [95% repository-bootstrap CI]; 5,000 replicates. Motivation and psychological safety are separate predictors; configuration indicators are included.

Predictor	Acceptance	Willingness	Behav. intention
Perceived quality	0.19 [−0.03, 0.41]	0.03 [−0.23, 0.32]	0.10 [−0.14, 0.36]
Perceived utility	0.36 [0.20, 0.51]	0.14 [−0.06, 0.25]	0.21 [0.06, 0.35]
Trust	0.24 [0.09, 0.42]	0.06 [−0.08, 0.22]	0.13 [−0.14, 0.40]
Motivation	−0.09 [−0.20, 0.03]	0.27 [0.03, 0.46]	−0.05 [−0.20, 0.10]
Psych. safety	−0.05 [−0.12, 0.00]	−0.03 [−0.17, 0.07]	0.07 [−0.09, 0.25]

below 7.3% on all fourteen items. On the reverse-worded item, 48.8% rejected the idea of arguing with the AI and 30.8% endorsed it.

The Acceptance item directly addresses agreement with the AI judgement (79.9% endorsement). Future-use intention is lower (68.0%), as is motivation (41.9%, with 53.2% neutral). These are instance-level response proportions, not percentages of independent students. High perceived quality therefore does not imply reported motivation or future use.

Conditional associations with acceptance. Table 3 reports within-student coefficients and repository-bootstrap intervals. Utility was positively associated with acceptance and future-use intention. Trust was positively associated with acceptance. Perceived-quality intervals included zero after configuration adjustment; without that adjustment its acceptance interval was [0.02, 0.44]. Motivation was positively associated with willingness; psychological-safety intervals included zero throughout. The fifteen exploratory intervals are not adjusted for multiplicity, and coefficient differences were not directly tested.

5.3 RQ2: Differences Between Feedback Configurations

Table 4 answers the direct agreement comparison: all six Acceptance intervals include zero. This uncertainty neither demonstrates equal agreement nor identifies a preferred

Table 4. All six paired configuration contrasts in the survey export. Mean difference [repository-bootstrap CI]; 5,000 draws. Positive values favour the first configuration. Acceptance is the direct agreement item (95% CI); AAI is a secondary multidomain composite (90% CI for exploratory margin assessment). Intervals are not multiplicity-adjusted.

Contrast	Acceptance: difference [95% CI]	AAI: difference [90% CI]
Sandwich–Rubric	0.047 [-0.121, 0.185]	0.114 [0.008, 0.212]
Sandwich–Explainable	-0.012 [-0.165, 0.122]	0.056 [-0.028, 0.128]
Sandwich–Researching	-0.023 [-0.198, 0.136]	0.029 [-0.066, 0.110]
Rubric–Explainable	-0.058 [-0.204, 0.084]	-0.058 [-0.119, -0.003]
Rubric–Researching	-0.070 [-0.274, 0.120]	-0.085 [-0.162, -0.014]
Explainable–Researching	-0.012 [-0.176, 0.153]	-0.027 [-0.076, 0.025]

configuration. Secondary AAI means range from 4.01 (Rubric) to 4.12 (Sandwich), while individual outcome profiles differ (Fig. 2). A secondary Friedman analysis gives $\chi^2(3) = 23.04$, $p < .001$, $W = .089$; a student-intercept mixed-model omnibus comparison gives $p = .084$. Neither test fully models shared-repository dependence, so neither is the principal basis for the configuration interpretation.

The largest profile differences concern utility and actionability. Utility means are 4.27 for Sandwich, 4.22 for Researching, 4.01 for Rubric and 3.95 for Explainable. Actionability means are 4.33, 4.28, 3.97 and 3.95, respectively. Sandwich–Explainable actionability differs by 0.37 points (95% repository-bootstrap CI [0.16, 0.57]). Deleting each repository in turn gives Sandwich–Explainable actionability point estimates of 0.34–0.41, with no sign reversal across the 28 deletions. These deletion ranges assess sensitivity to a single repository; they are not confidence intervals. The contrast compares complete configurations and does not identify a next-steps effect. Trust is descriptively lowest for Rubric (3.91 versus 4.11–4.12); Researching has lower mean clarity (4.19 versus 4.36–4.41). These item profiles should not be replaced by the small differences in the multidomain average.

Researching feedback is longer: 623 whitespace-separated text units on average versus 215–267 for the other configurations. A secondary student-intercept model adjusting for feedback and answer length retains an estimated Rubric–Sandwich AAI gap near -0.11 points, while the length-covariate intervals include zero. Feedback length is generated by the configuration, so adjustment neither identifies a direct configuration effect nor rules out a role for length. This proxy counts strings in stored JSON, excluding identifiers, configuration labels and model names; it is neither Vietnamese lexical word count nor provider token usage. Cost comparisons require usage logs.

All six repository-bootstrap 90% intervals for paired AAI mean differences were inside the exploratory ± 0.25 margin; Sandwich–Rubric was $[0.008, 0.212]$. This supports only small average composite differences under that chosen margin, not equal item profiles or a validated practical-equivalence threshold. Margin sensitivity is consequential: only five of six intervals fit inside ± 0.20, and one fits inside ± 0.10. These are descriptive threshold checks, not calibrated educational equivalence claims. Retaining all fifteen items with Rejection reverse-scored gave a mean of 4.01 and again placed all six intervals inside the margin. The Friedman result persisted after excluding straight-lined rows (64 complete students, $\chi^2(3) = 23.51$, $W = .12$) and after excluding the single

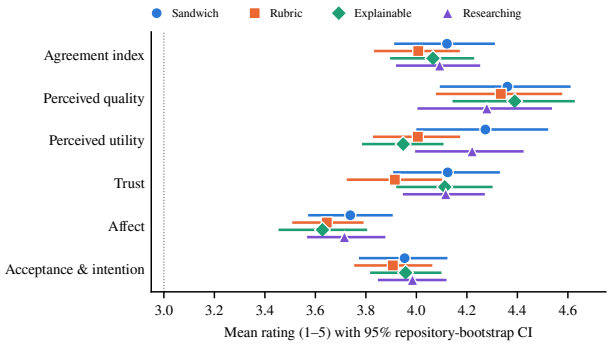


Fig. 2. Agreement index and construct scores by feedback configuration, with 95% repository-bootstrap confidence intervals (28 clusters). The dotted line marks the neutral midpoint.

Table 5. Agreement by question context type, mean. b: within-student estimate for raw code vs. graph, controlling for configuration.

Context	n	AAI	Quality	Utility	Trust	Acceptance & int.
Graph	280	4.06	4.30	4.11	4.05	3.94
Raw code	64	4.14	4.52	4.12	4.16	4.00
$b_{\text{raw}} = 0.07$, CI95% $[-0.02, 0.20]$; repository-bootstrap interval						

guardrail-refusal block (85 students, $\chi^2(3) = 23.22$). These response exclusions are sensitivity checks, not evidence that flagged responses are invalid.

In the intercept-only model, most AAI variance lay between students: the intraclass correlation of the agreement index was .77 (also .77 without straight-lined responses), indicating substantial clustering by student. This variance decomposition does not identify why configuration differences are small.

5.4 RQ3: Raw-Code versus Graph-Based Questions

Mean AAI was 4.14 for raw-code and 4.06 for graph-based questions (Table 5). The student fixed-effects model with configuration adjustment estimated $b = 0.07$ (95% repository-bootstrap CI $[-0.02, 0.20]$). The 26 students exposed to both types belong to 12 repositories; these supply the within-student context variation. Deleting each repository in turn gives raw-code coefficients of 0.047–0.114, with all 28 fits full rank; this point-estimate range is not a confidence interval and does not remove routing confounding. Among those 26 students, pooled means were 3.99 and 4.01, respectively; these are descriptive rather than a balanced paired estimate. Perceived quality averaged 4.52 versus 4.30, and transparency 4.59 versus 4.22. The routing rule ties context to method count, and the same context type feeds the evaluator. These estimates cannot separate complexity, question generation and feedback context; the interval crossing zero does not establish equivalence.

6 Discussion

Acceptance, motivation and reported use. In the survey export, endorsement of accuracy and usefulness exceeds endorsement of motivation and future use. Utility was positively associated with acceptance and intention within students; trust was associated with acceptance, consistent with technology-acceptance research [5,34]. Motivation was associated with willingness when separated from psychological safety. These self-reports do not establish actual revision or correctness. Appropriate reliance still requires independent comparison with actual feedback reliability [23].

Configuration bundles motivate controlled tests. Small AAI differences coexisted with an actionability gap of about 0.37 points between Sandwich and Explainable. The configurations designed to include next steps had higher mean actionability, but content, scoring, length and model orchestration also differed. An experiment manipulating next steps independently is needed. Researching produced more text without a detected AAI advantage over Sandwich; efficiency claims require usage logs. The high student ICC supports accounting for individual differences in comparisons.

Response patterns and measurement. Each student supplied sixty ratings; 18.6% of response sets were straight-lined. Response burden and reverse wording warrant scrutiny [21,33]. A shorter instrument, attention checks and behavioural probes merit evaluation; straight-lining alone does not establish inattention.

Threats to validity. Internal validity. Each configuration evaluates a different answer; question difficulty, order and carryover remain possible influences. Bundle comparisons cannot isolate next steps or model effects. Human completion is author-confirmed, not independently authenticated against database history; exported model labels likewise do not verify provider execution. *Construct validity.* The instrument lacks factorial validation. AAI combines domains with unequal total weights, and excluding Rejection was data-informed. Pooled reliability mixes within- and between-student covariance; the affect pair has low consistency. Reliability and human-comparison items ask about experiences that this design cannot independently verify. Concurrent self-reports may share common-method variance [30]. No expert correctness, observed revision or learning outcome was measured. *External validity.* Voluntary classroom participation at one university, defence preparation and novelty limit transfer. Random class selection does not remove respondent self-selection. RQ3 couples context with method count and evaluator context. *Statistical conclusion validity.* Repository resampling preserves nested student and question dependence, assuming independent repositories. With only 28 clusters, percentile-bootstrap coverage is approximate. Rank tests and student-intercept models omit some repository dependence and remain secondary. Deletion ranges are sensitivity summaries, not CIs. Regression associations are contemporaneous, and their fifteen intervals are unadjusted for multiplicity. High ratings may limit discrimination. The uncalibrated ±0.25 AAI margin establishes no item-level or educational equivalence. Neither response-pattern exclusions nor straight-lining establish response invalidity.

7 Conclusion

PACE connects repository analysis, scenario questions and structured feedback in a classroom comprehension workflow. Across 344 rated instances from 86 students, endorsement of accuracy and usefulness exceeded endorsement of motivation and future use. Agreement with the AI judgement was measured directly by the Acceptance item: its six configuration contrasts all included zero, without demonstrating equivalence. Small differences in a secondary multidomain composite coexisted with a larger actionability difference between Sandwich and Explainable. The bundle comparison does not identify next-steps or model effects, and the context comparison remains observational.

The contribution is a field evaluation of reported perceptions in a team-repository setting. Future work should validate the instrument, audit technical correctness and observed revisions, and manipulate feedback features independently. Deployment logs, research consent and data-governance documentation remain necessary for a complete submission record.

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Research ethics and data governance Author confirmation required before submission: classroom recruitment is described in Section 4. The authors affirm voluntary participation, including non-participation by some classmates. This account does not by itself document consent for research use, grading relationships, institutional review or exemption, or permission to process repository content through external APIs; supporting records and safeguards remain to be documented. No approval or consent procedure is asserted without supporting records.

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